CISS362: Introduction to Automata Theory, Languages, and Computation
Assignment 2

You should study Sipser chapter 0 and do all the problems in the Exercises and Problems section. Form study groups. Meet at CS Hangout. Compare solutions. As always I very strongly encourage discussion. But you must must write your own work and you must understand what you wrote down. If I call on you to explain your work and you can't, then that's considered plaigiarism.

For Q1, enter your work in `q01.tex`. Several examples are included so that you and modify the given LaTeX code. You can also find some information on LaTeX at my website `http://yliow.github.io` (go to the Tutorial section and look for the latex pdf.) Or just ask me.

Proving $P(n)$ for all $n \geq n_0$ where n_0 is a fixed integer

Recall that if you want to prove $P(n)$ is true for all $n \geq 42$, you have to do two things:

- Prove $P(42)$ is true
- Let $n \geq 42$. Assume $P(n)$ holds and prove $P(n+1)$ holds as well.

If you can achieve the above two points, then you can claim

- $P(n)$ is true for all $n \geq 42$

This is one form of mathematical induction called **weak mathematical induction**. The **strong mathematical induction** says that if you can do two things:

- Prove $P(42)$
- Let $n \geq 42$. Assume $P(42), P(42+1), ..., P(n)$ holds and prove $P(n+1)$ holds as well.

If you achieve the above two points, then you can claim

- $P(n)$ is true for all $n \geq 42$

The "42" above can be replaced by any integer (including some negative integer).

The above two induction techniques proves $P(n)$ are all true for $n \geq n_0$. Mathematical induction also allows you go "go backward". If you can do the following:

- Prove $P(n_0)$ is true
- Let $n \leq n_0$. Assume $P(n)$ holds and prove $P(n-1)$ holds as well.

then you can claim

- $P(n)$ is true for all $n \leq n_0$

This is also called weak mathematical induction. The strong induction going backward holds too. If you can achieve the following:

- Prove $P(n_0)$ is true
- Let $n \leq n_0$. Assume $P(n_0), P(n_0-1), ..., P(n)$ holds and prove $P(n-1)$ holds as well.

then you can claim

- $P(n)$ is true for all $n \leq n_0$

So induction allows you to prove $P(n)$ when n goes to infinity or when n goes to negative infinity.

There is another method that's important and does the same thing for you.

Sometimes, it is possible to prove $P(n)$ for all $n \geq 1$ by using the "proof by contradiction" method by using the well-ordering principle (WOP) which says that

- Every nonempty subset of $\mathbb{N} = \{0, 1, 2, ...\}$ has a least element, i.e., if X is a nonempty subset of $\mathbb{N}$, then there is some $m \in X$ such that $m \leq x$ for all $x \in X$.

(See class notes.) In this case, frequently, the proof involves an argument of the form:

"Suppose it's not true that $P(n)$ holds for all $n \geq n_0$". Then the set

$$X = \{n \mid n \geq n_0,\ \ P(n) \text{ does not hold}\}$$

is a nonempty subset of $\mathbb{N}$. By WOP, X has a least element, i.e., there is a smallest m such that $m \geq n_0$ and $P(m)$ is false. And you continue to prove that something goes wrong, i.e.,you attempt to arrive at a contradiction. This is frequently done in one of two ways. Here's one way to achieve this: Since m is the least element of X, all $n \in \mathbb{N}$ with $n < m$ must satisfy $P(n)$. From such n, you then show that in fact $P(m)$ holds, which clearly is a contradiction. The second method is this: Try to find some $m' < m$ such that $P(m')$ is also false. This would contradict the fact that m is the least elment of X.

(Aside. WOP is related to the proof method called **Fermat's infinite descent**. Applying the same argument above on k' and assuming $P(k')$ is false, you would arrive at another $k'' < k'$ such that $P(k'')$ is false, etc. This gives you infinitely many positive integers $k > k' > k'' > k''' > \cdots$. This is clearly impossible since there can only be finitely many positive integer from n_0 up to k.)

Until you know how to write induction proofs properly, you must follow these instructions for writing a proof that uses induction.

1. Paragraph 1: State you $P(n)$ and the range of values for n. If a problem involves proving multiple statements, you can also use $Q(n)$, $R(n)$, etc. State what method you are using (weak or strong induction). The default is weak induction, i.e., if you want you are using mathematical induction, it means you are using weak mathematical induction.
2. Paragraph 2: State you are proving the base case. Prove the base case.
3. Paragraph 3: State you are proving the inductive case. State your inductive hypothesis and state what you are going to prove. Then prove it. If the proof is long, state what you have proven. It's even a good idea to state it anyway. That's called good writing: State at the beginning of a paragraph what you want to do, do it, then remind the reader the goal at the beginning of the paragraph. (This is the longest part of the proof. If necessary, you might need more than one paragraph.)
4. Paragraph 4: State, quoting the method (i.e., induction), what you have proven.

Once you are done with writing about 50 induction proofs, you can use a freer form.

Here are some examples:

Theorem 0.1. *If $n \geq 0$, then*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

(Note that if $n = 0$, then the expression on the left-hand side of the above equation is 0 by definition. In other words an empty sum – sum of no terms – is defined to be 0.)

Proof. We will prove this by weak mathematical induction. For $n \geq 0$, we define

$$P(n) = \left(1 + 2 + \cdots + n = \frac{n(n+1)}{2}\right)$$

Base case. When $n = 0$, we have

$$1 + 2 + \cdots + n = 0 = \frac{0(0+1)}{2} = \frac{n(n+1)}{2}$$

Hence $P(0)$ holds.

Inductive case. Assume $P(n)$ holds where $n \geq 0$, i.e., we assume

$$P(n) = \left(1 + 2 + \cdots n = \frac{n(n+1)}{2}\right)$$

holds. We want to show $P(n+1)$ holds, i.e., we want to show

$$P(n+1) = \left(1 + 2 + \cdots n + 1 = \frac{(n+1)(n+1+1)}{2}\right)$$

is true. Since $P(n)$ holds, we have

$$\begin{aligned} 1 + 2 + \cdots + n &= \frac{n(n+1)}{2} \\ \therefore \quad 1 + 2 + \cdots n + (n+1) &= \frac{n(n+1)}{2} + (n+1) \\ &= \frac{n(n+1) + 2(n+1)}{2} \\ &= \frac{(n+1)(n+2)}{2} \\ &= \frac{(n+1)((n+1)+1)}{2} \end{aligned}$$

i.e., $P(n+1)$ holds.

Therefore, by weak mathematical induction, $P(n)$ holds for all $n \geq 0$, i.e., for all $n \geq 0$,

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

□

Theorem 0.2. *If $n \geq 0$, then*

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$$

Proof. We will prove this using the well-ordering principle. Let $P(n)$ be the statement

$$P(n) = \left(1 + 2 + \cdots + n = \frac{n(n+1)}{2}\right)$$

Suppose on the contrary that $P(n)$ does not hold for all $n \geq 0$. Let $n_0 \in \mathbb{N} = \{0, 1, 2, ...\}$ such that $P(n_0)$ does not hold. Note that by definition when $n = 0$,

$$1 + 2 + \cdots + n = 0 = \frac{0(0+1)}{2} = \frac{n(n+1)}{2}$$

Hence the statement $P(0)$ holds. Therefore $n_0 > 0$.

Let

$$X = \{n \mid n \geq 0, \;\; P(n) \text{ does not hold}\}$$

Note that $X \subseteq \mathbb{N} = \{0, 1, 2, ...\}$. Since $n_0 \in X$, X is a nonempty subset of $\mathbb{N} = \{0, 1, 2, ...\}$. Therefore by the well-ordering principle, X has a least element, say m. Since $m \in X$, we have

$$1 + 2 + \cdots + m \neq \frac{m(m+1)}{2} \tag{a}$$

i.e., $P(m)$ does not hold. Since $P(0)$ holds, we have $m > 0$ and hence $m - 1 \geq 0$. Since m is the least element of X, $m - 1 \geq 0$ is in $\mathbb{N} = \{0, 1, 2, ...\}$, we have $m - 1 \notin X$. Therefore

$P(m-1)$ holds, i.e.,

$$\begin{aligned}
1+2+\cdots+(m-1) &= \frac{(m-1)((m-1)+1)}{2} \\
\therefore\quad 1+2+\cdots+(m-1)+m &= \frac{(m-1)((m-1)+1)}{2} + m \\
&= \frac{(m-1)m}{2} + m \\
&= \frac{(m-1)m+2m}{2} \\
&= \frac{m^2-m+2m}{2} \\
&= \frac{m^2+m}{2} \\
&= \frac{m(m+1)}{2}
\end{aligned}$$

Hence $P(m)$ holds. This contradicts (a). Therefore X must be empty. Hence

$$1+2+\cdots+n = \frac{n(n+1)}{2}$$

for all $n \geq 0$. □

Theorem 0.3. *Let T be a tree wth at least one node, i.e., a connected simple graph with at least one node and no cycles. Then $e = v - 1$ where e and v are the number of edges and nodes of T respectively.*

(Note: A simple graph is a graph with no loops, i.e., no edge joining a node to itself and no multi-edges, i.e., no multiple edges joining the same two nodes. A graph is connected is for every pair of distinct vertices x, y, there is a path of edges from x to y.)

Proof. For a graph G, we will write v_G and e_G for the number of nodes and number of edges of G (respectively). For $n \geq 0$, we define the proposition $P(n)$ as follows:

$$P(n) = (\text{If } T \text{ is a tree with at least one node and with } n \text{ edges, then } e_T = v_T - 1)$$

We will prove $P(n)$ holds for all $n \geq 0$ by strong mathematical induction.

Base case. We will prove $P(0)$, i.e., we will prove that if T is a tree with at least one node and 0 edges (i.e., $e_T = 0$), then $e_T = v_T - 1$. If T has at least two distinct nodes, say x and y, then x and y are not adjacent since there are no edges in T. But T is connected. This is a contradiction. Hence T has exactly one node, i.e., $v_T = 1$. Therefore

$$e_T = 0 = 1 - 1 = v_T - 1$$

i.e., $P(0)$ holds.

Inductive case. Let $n \geq 0$. Assume $P(0), P(1), ..., P(n)$ holds. We will show that $P(n+1)$ holds, i.e., we will show that if T is a tree with at least one node and has $n + 1$ edges, then

$$e_T = v_T - 1$$

Let T be a tree with at least one node and $n + 1$ edges. Since $n \geq 0$, we have $n + 1 \geq 1$. Hence there is at least one edge in T, say e denote an edge in T joining node x and node y. Since T is simple, $x \neq y$. Construct the graph

$$G = T - e$$

i.e., G is the graph T with edge e removed. We have

$$v_G = v_T$$
$$e_G = e_T - 1$$

Note that G contains all the nodes of T and in particular contains x and y. We claim that G is made up of two disjoint trees.

First, we show that G has exactly two connected components, i.e., G is made up of two

maximal connected subgraphs. Suppose k is the number of connected components of G. Let $G_1, G_2, ..., G_k$ denote the connected components of G. There are no paths joining a node in G_i to a node in G_j if $i \neq j$. We will show $k = 2$ and furthermore each G_i is a tree.

The nodes x and y are in G. Therefore x is in some G_i and y is in some G_j. Note that $i \neq j$. Otherwise, if $i = j$, x and y are in the same connected component G_i which implies that there is some path p in G_i joining x and y. Since G_i is a subgraph of $G = T - e$ and G does not contain edge e, G_i cannot contain e. The path p (which is in G_i) therefore also cannot contain edge e. Hence p and edge e will form a cycle in T. This is a contradiction since T is a tree and cannot have a cycle. Hence we have shown that $i \neq j$, i.e., x and y are in two different connnected components of G.

Now suppose $k > 2$, i.e., suppose there are three connected components. Recall that x is in some G_i and y is in some G_j with $i \neq j$. Since there are at least three connected components, there is some k such that $k \neq i, k \neq j$. Let z be a node in G_k. Since T is a tree, there is a path p in T from x to z. There are no repeated nodes in p. Since x and z are in different connected components, the path p must leave G_i and must enter G_k. The only edge that leaves G_i is the edge e. Since y is in G_j, path p will leave G_i and enter G_j. Therefore on entering G_j, p contains x and y. However to arrive at G_k, which is not G_j, the path p has to leave G_j. The only edge leaving G_j is e, which means that the path p on leaving G_k will repeat x. This is a contradiction. We conclude that $k = 2$, i.e., there are two connected components in $G = T - e$.

We have now shown that there are two connected components G_1, G_2 in G.

We now show that the connected components G_1, G_2 in G are trees. Suppose there is a cycle C in G_1. Since G_1 is in $G = T - e$, the cycle C is also in G and is therefore in T. This is a contradiction since T is a tree. (This shows that every subgraph of a tree cannot have cycles.)

We have now shown that $T - e$ is made up of two disjoint trees, say T_1 and T_2.

Since T_1, T_2 are both subgraphs of G and G is a subgraph of T,

$$e_{T_i} \leq e_G = e_T - 1 < e_T = n + 1$$

for $i = 1, 2$. Therefore by induction hypothesis,

$$e_{T_1} = v_{T_1} - 1$$
$$e_{T_2} = v_{T_2} - 1$$

Adding these two equations, we get

$$\begin{aligned} e_{T_1} + e_{T_2} &= v_{T_1} - 1 + v_{T_2} - 1 \\ \therefore \quad e_{T_1} + e_{T_2} &= v_{T_1} + v_{T_2} - 2 \end{aligned} \tag{a}$$

Now note that since $T = G - e$, the only difference between T and G is an edge. Hence

$$e_T = e_G + 1 = e_{T_1} + e_{T_2} + 1 \tag{b}$$

$$v_T = v_G = v_{T_1} + v_{T_2} \tag{c}$$

Hence equations (a), (b), (c) gives us

$$\begin{aligned} e_T &= e_{T_1} + e_{T_2} + 1 \\ &= (v_{T_1} - 1) + (v_{T_2} - 1) + 1 \\ &= v_{T_1} + v_{T_2} - 1 \\ &= v_T - 1 \end{aligned}$$

We have now shown $P(n+1)$ holds if $P(0), P(1), ..., P(n)$ hold.

Therefore, by strong mathematical induction, $P(n)$ holds for all $n \geq 0$, i.e., we have shown that if T is a tree with at least one node then

$$e_T = v_T - 1$$

□

Template for induction proofs

A. Here is the template for weak induction proofs:

We will prove the above statemet by weak induction. For $n \geq ?$, let $P(n)$ be the statements

$$P(n) = \Big(n^2 \text{ is a prime}\Big)$$

Base case. We now prove the base case $P(?)$. [... your proof ...] Hence $P(?)$ holds.

Inductive case. We now prove the inductive case. Assume $P(n)$ holds where $n \geq ?$. [... your proof ...] Hence $P(n+1)$ holds.

Therefore, by weak mathematical induction, $P(n)$ is true for all $n \geq ?$, i.e., for any $n \geq ?$,

$$n^2 \text{ is a prime}$$

B. Here is the template for strong induction proofs:

We will prove $P(n)$ is true for all $n \geq ?$ using strong mathematical induction. For $n \geq ?$, let $P(n)$ be the statements

$$P(n) = \Big(n^2 \text{ is a prime}\Big)$$

Base case. We now prove the base case $P(?)$. [... your proof ...] Hence $P(?)$ holds.

Inductive case. We now prove the inductive case. Assume $P(?), P(?+1), ..., P(n)$ hold where $n \geq ?$. [...] Hence $P(n+1)$ holds.

Therefore, by strong mathematical induction, $P(n)$ is true for all $n \geq ?$, i.e., for any $n \geq ?$,

$$n^2 \text{ is a prime}$$

B. Here is a template for WOP proofs:

We will prove the above statement using the well-ordering principle. For $n \geq ?$, let $P(n)$ be

$$P(n) = \Big(n^2 \text{ is a prime}\Big)$$

Assume on the contrary that $P(n)$ does not hold for all $n \geq ?$. Then there is some n such

that $P(n)$ does not hold. Define

$$X = \{n \geq ? \mid P(n) \text{ does not hold}\}$$

[... now prove that X is a nonempty subset of $\mathbb{Z}$ that is bounded below, or a nonempty subset of $\mathbb{N}$.]

By the well-ordering principle X has a least element, say m. Since m is the least element of X, $P(n)$ holds for all $? \leq n < m$. [Now derive a contradiction. Warning: You usually have to use some n such that $? \leq n < m$. This means that you have to explain why $m >?$.] This is a contradiction because [...]

Therefore, $P(n)$ holds for all $n \geq ?$.

Q1. Sipser 0.10.

Find the error in the following proof that $2 = 1$.

Consider the equation $a = b$. Multiply oth sides by a to obtain $a^2 = ab$. Subtract b^2 from both sides to get $a^2 - b^2 = ab - b^2$. Now fator each side, $(a + b)(a - b) = b(a - b)$, and divide each side by $(a - b)$ to get $a + b = b$. Finall, let a and b equal 1, which shows that $2 = 1$.

SOLUTION.

The error occurs when dividing both sides by (a - b). If a and b are supposed to be set equal to each other, then subtracting them would cancel out and cause a division of zero, which is undefined.

Q2. Sipser 0.12.

SOLUTION.

The proof claims that every set of k horses have the same color. It removes one horse to obtain H1 and a different horse to obtain H2. The proof claims that because all horses in H1 are the same color and all horses in H2 are the same color, then all horses in H must have the same color.

The problem when k = 1:

When k = 1, the set H = k + 1 = 2 horses.

Let's say we have 2 horses: Horse A - Brown Horse B - White

We remove Horse A to obtain H1 Horse B

By the induction process, all horses in H1 have the same color

Now we remove Horse B to obtain H2 Horse A

Again, all horses in H2 have the same color

However, H1 and H2 have no horses in common. And the proof cannot conclude that the horses in H1 and H2 are the same.

Q3. Using mathematical induction, give a complete proof of the following fact:

$$4(1^3 + 2^3 + \cdots + n^3) = n^2(n+1)^2$$

for $n \geq 0$.

SOLUTION.

We will prove the above statement by weak induction. For n ¿= 0, let P(n) be the statments

P(n) = $4(1^3 + 2^3 + ... + n^3) = n^2(n+1)^2$

Basis: We now prove the base case P(0). P(0): $4(0) = 0(0)$ -¿ $0 = 0$ Hence P(n) holds

Induction Step: We now prove the inductive case.

For each k ¿= 1, assume the claim is true for P(k) and prove that it is true for p(k + 1). The induction hypothesis states that

$4(1^3 + 2^3 + \cdots + k^3) = k^2(k+1)^2$

Our objective is to prove that

$4(1^3 + 2^3 + \cdots + k^3 + (k+1)^3) = (k+1)^2(k+2)^2$

Starting on the left-hand side:

$4(1^3 + 2^3 + \cdots + k^3 + (k+1)^3) = 4(1^3 + 2^3 + \cdots + k^3) + 4(k+1)^3 = k^2(k+1)^2 + 4(k+1)^3 = (k+1)^2[k^2+4(k+1)] = (k+1)^2+[k^2+4k+4] = (k+1)^2+[(k+2)(k+2)] = (k+1)^2+(k+2)^2$

Hence P(k + 1) holds

Therefore, by weak mathematical induction, P(n) is ture for all n ¿= 0. i.e., for any n ¿= 0.

$4(1^3 + 2^3 + ... + n^3) = n^2(n+1)^2$

Q4. Using WOP, give a complete proof of the following fact:

$$4(1^3 + 2^3 + \cdots + n^3) = n^2(n+1)^2$$

for $n \geq 0$.

SOLUTION.

We will prove the above statement using the well-ordering principle. For n ¿= 0. P(n) be the statements

$4(1^3 + 2^3 + ... + n^3) = n^2(n+1)^2$

Assume on the contrary that P(n) does not hold for all n ¿= 0. Then there is some n such that P(n) does not hold. Define

S = n Z¿=0 : P(n) is false is nonempty. By the WOP, S has a least element, say m. Since k is the least element of S, P(n) holds for all 0 ¡= n ¡ m.

Case 1: m = 0 When n = 0, the sum is empty, so its value is 0. Therefore, $4(1^3+2^3+...+0^3) = 0 and 0^2(0+1)^2 Thus P(0) holds, contradicting the fact that m is a counterexample Therefore, m > 0$

Case 2: m ¿ 0 Since m is the smallest counterexample, m - 1 cannot be a counterexample. Therefore P(m - 1) is true. Consider P(m): $4(1^3 + 2^3 + ... + m^3) = 4(1^3 + 2^3 + ... + (m-1)^3) + 4m^3 = (m-1)^2m^2 + 4m^3$

(m - 1)$^2m^2 + 4m^3 = m^2((m-1)^2 + 4m) = m^2(m^2 - 2m + 1 + 4m) = m^2(m^2 + 2m + 1) = m^2(m^2+1)^2$

Therefore, $4(1^3+2^3+...+m^3) = m^2(m+1)^2 Thus, P(m) is true, contradicting the fact that m is a counterexa$

This is a contradiciton because we show that there are no counterexample, after assuming counterexamples exist

Therefore, P(n) holds for all n ¿= 0.

Q5. Write a complete proof of the Euclidean property of integers: If a and b are integers such that $a \geq 0$ and $b > 0$, then there exists integers q and r such that $q \geq 0$ and

$$a = bq + r, \;\; 0 \leq r < b$$

You must use WOP. (I have given you the proof in class. The point of this question is to study the proof in depth, write it up properly in your own words, and edit your work until it's perfect.)

SOLUTION.

Write a complete proof of the Euclidean property of integers: If a and b are integers such that a 0 and b ¿ 0, then there exists integers q and r such that q 0 and a = bq + r, 0 r ¡ b

We will prove the above statement use the well-ordering principle. For a ¿= 0, b ¿ 0

a = bq + r, for some q ¿= 0, and 0 ¡= r ¡ b

Consider the set

X = a - bx — x N, a - bx ¿ 0

This set is a nonempty set because x = 0 gives

a - b(0) = a ¿= 0

Thus a X

Since X is a nonempty set with non-negative numbers, the WOP says that X has a least element. Let this least element be r. Therefore, for some integer x ¿= 0

r = a - bx

rearanging gives

a = bx + r

also since r X

r ¿= 0

Now, I am going to show that r ¡ b. Assume on the contrary that r ¿= b, then

r - b ¿= 0

but r - b = (a - bx) - b = a - b(x + 1)

since x ¿= 0 we have

x + 1 ¿= 0

Therefore, r - b X. However, r - b ¡ r, because b ¿ 0. This is a contradiction because of the fact r was the least element of X. So, our assumption was false and r ¡ b.

Combining everything, we have found integers q, r such that

q ¿= 0, a = bq + r, 0 ¡= r ¡ b

Therefore, by the WOP, the Euclidean property of integers holds

Primes

Note that a number n is said to be a **prime** if it is a whole number that can only be divided by 1 and itself (i.e., n). The positive integer n is said to be **composite** if it is greater than 1 and is not a prime, which means that it is possible to write n as a product, $n = a \cdot b$, where a and b are positive integers such that $1 < a < n$ and $1 < b < n$. A positive integer n (positive means > 0) must fall into exactly one of the 3 cases:

- n is 1
- n is prime
- n is composite.

Note that if a and b are integers and $a > 0$, $b > 0$, then $a \leq ab$ and $b \leq ab$. And if $a > 1$, then $b < ab$.

Q6. Consider the following fact: Every positive integer $n \geq 1$ is a product of primes. For instance for $n = 20$, the ordered collection of primes involved are $(2, 2, 5)$ (in ascending order). We define the product of the empty collection of primes, i.e. (), to be 1. Of course the collection can have one single number: If the collection is (11), then the product is 11.

Define $P(n)$ to be the above statement, i.e.,

$$P(n) = \Big(\text{Every positive integer } n \geq 1 \text{ is a product of primes}\Big)$$

Give a complete proof of $P(n)$ for all $n \geq 1$ using strong or weak induction. Here, I'm giving you the $P(n)$ for free.

You can assume Euclid's lemma: If p is a prime dividing ab, then p divides a or p divides b.

(In fact not only is every positive integer n representable as a product of primes, it is represented as a product of primes in a *unique way*. In other words, the ordered collection of primes for n is unique. This can also be proven using induction. But you don't have to prove it.)

SOLUTION.

We are going to prove the above statement using weak induction. For n ¿= 1, we will prove that every integer n is a product of primes

Basis: We will now prove the base case P(1)

By definition, the product of the empty collection of primes () is 1. Therefore 1 is a product of primes. Thus, P(1) holds

Induction Step: We now prove the unductive case.

Assume every positive integer m with 1 ¡= m ¡ k is a product of primes. For each k ¿= 1, assume the claim is true for P(k) and prove that it is true for P(k + 1).

There are 2 cases

Case 1: k + 1 is prime

Then k + 1 is already a product of primes consisting of a one-element collection, (k + 1)

Therefore P(k + 1) is true

Case 2: k + 1 is not prime

Since k + 1 ¿ 1 is not prime, it is composite. Therefore, there exists integers ab such that k + 1 = ab. With 1 ¡ a ¡ k + 1, and 1 ¡ b ¡ k + 1. In particular a ¡= k and b ¡= k.

By the induction hypothesis, every positive integer from 1 to k is a product of primes. Therefore, both a and b are products of primes.

Suppose a = $p_1 p_2 ... p_i and b = q_1 q_2 ... q_j Where all p_i and q_j are primes$

Then k + 1 = ab = $(p_1 p_2 ... p_i)(q_1 q_2 ... q_j) = p_1 p_2 ... p_i q_1 q_2 ... q_3$

Thus k + 1 is also a product of primes and P(k + 1) holds

Therefore, by weak mathematical induction, P(n) is true for all n ¿= 1. i.e., for any n ¿= 1. P(n) = (Every positive integer n ¿= 1 is a product of primes)

Q7. Using mathematical induction, prove that for $n \geq 4$, $2^n \geq n^2$.

(Actually this can also be proven using Calculus.)

SOLUTION.

We are going to prove the above statement using weak induction. For n ¿= 4, let P(n) be the statement

$2^n >= n^2$

Basis Step: We will now prove the base case P(4)

$P(4) = 2^4 >= 4^2 = 16 >= 16$

Thus, P(4) holds

Induction Step: We will now prove the inductive case

For each k ¿= 1, assume the claim is true for P(k) and prove that is is true for P(k + 1). The induction hypothesis states that

$2^k > k^2, for k >= 4$

Our object is to prove that

$2^(k+1) >= (k+1)^2$

So,

$2^(k+1) = 2 * 2k^2 >= 2k^2 >= (k+1)^2$

Hence, P(k + 1) holds

Therefore, by weak mathematical induction, P(n) is true for all n ¿= 4.

$2^n >= n^2$

Q8. Prove or disprove the following: Let X and Y be sets.

(a) $P(X \times Y) = P(X) \times P(Y)$
(b) $P(X \cup Y) = P(X) \cup P(Y)$
(c) $P(X) = 2^{|X|}$ if X is a finite set.

Remember that to disprove a statement, all you need is a counterexample. When constructing examples, always construct the simplest.

SOLUTION.

(a) P(X $\times Y) = P(X) \times P(Y) This statement is false Counterexample$:

X = 1, Y = 2

P(X $\times Y) = \emptyset, (1, 2)$

P(X) $= \emptyset, 1 P(Y) = \emptyset, 2$

Therefore,

P(X)$\times P(Y) = \emptyset, (\emptyset, 1), (\emptyset, 2), (1, 2)$

—P(X$\times Y)| = 2|P(X) \times P(Y)| = 4$

Thus, P(X$\times Y$)! $= P(X) \times P(Y) and (a) is false$

(b) P(X$\cup Y) = P(X) \cup P(Y) This statement is false Counterexample$:

X = 1, Y = 2

X$\cup Y = 1, 2$

P(X$\cup Y) = \emptyset, 1, 2, (1, 2)$

P(X) $= \emptyset, 1 P(Y) = \emptyset, 2$

Therefore,

P(X)$\cup P(Y) = \emptyset, 1, 2$

Now 1,2$\in P(X \cup Y), but 1, 2 \notin P(X) \cup P(Y)$.

Therefore, P(X$\cup Y$)! $= P(X) \cup P(Y) and (b) is false$

(c) $|P(X)| = 2^{|X|}$ if X is a finite set

This statment is true

Suppose

$|X| = n$

To construct a subset of X, we make a decision for each element of X, include it or don't include it. So each element of n gives us to choices. By the the mulltiplication principle, the total number of subsets is

$2 * 2 * ... * 2 = 2^n$

Since $n = |X|$, the above statment is true.

Q9. Using mathematical induction, prove that an undirected simple graph with n nodes can have at most $n(n+1)/2$ edges. (Simple means there are no loop edges and no multiedges. A loop edge is an edge that joins the same node. Two edges are multiedges is the join the same two nodes.) $P(n)$ is the statement "if G is an undirected simple graph with n nodes, then the G has at most $n(n+1)/2$ edges".

SOLUTION.

We are going to prove the above statement using weak induction. For n ¿= 1, let P(n) be the statement

n(n + 1) / 2

Basis Step: We will now prove the base case P(1)

P(1) = 1(1 + 1)/2 = 1

Since G is a simple undirected graph, G with 1 node has 0 edges. Since 1 is the upper bound and 0 ¡= 1, P(1) holds.

Induction Step: We will now prove the inductive case

For each k ¿= 1, assume the claim is true for P(k) and prove that it is ture for P(k + 1). The induction hypothesis states that

A simple graph with k + 1 nodes has at most (k + 1)(k + 2)/2 edges

So, take some graph G with k + 1 nodes. Choose one node v, and remove it, and all edges connected to v. We are left with a simple graph containing k edges. By our induction, the graph has at most

k(k + 1)/2 edges

Now put v back. Because the graph is simple v can't have any edges to itself or multiple edges. Since there are k other edges, v can add at most k edges.

Simplify (k(k + 1)/2) + k

(k(k + 1)/2) + k = (k(k + 1) + 2k)/2 = $(k^2 + 3k)/2$

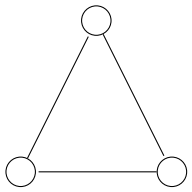
Our desired upper bound is (k + 1)(k + 2)/2 = $(k^2 + 3k + 2)/2$

Clearly $(k^2 + 3k)/2$ <= $(k^2 + 3k + 2)/2$

Hence, E(G) ¡= $(k^2 + 3k + 2)/2, and P(k + 1) holds$

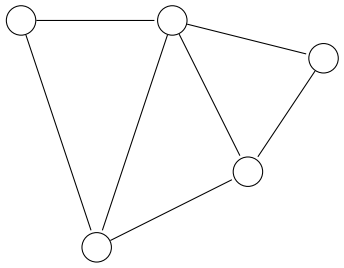
Therefore, by weak mathematical induction, P(n) is true for all n ¿= 1

Q10. A triangulation graph is a graph which looks like a patchwork of triangles, i.e., the graph is constructed by "gluing" triangles on a tabletop.



along edges. (Look at `main.tex` to see how to draw the above graph.)

Note that line (edge) crossings are not allowed. For instance here's one with 5 nodes (i.e. circles) and 7 edges (i.e. lines):



(a) Is it true that using the above construction (i.e., glueing triangle graphs along edges), all the "pieces" (i.e. finite regions) that you see are always triangles (i.e. a shape that is bounded by exactly 3 edges)? For instance in the above graph, there are 3 triangles.

(b) The degree of a node is just the number of edges (lines) joined to it. For the above type of graphs is it possible to draw one with exactly one node of degree 10? Show that it is possible by drawing such a graph or prove it's not possible.

(c) Is it possible to have a graph with 5 nodes of degree 2? Show that it is possible by drawing such a graph or prove it's not possible.

(d) Is it possible to have a graph with no nodes of degree 2? Show that it is possible by drawing such a graph or prove it's not possible.

(e) The degrees of the nodes in the above example is 2, 2, 3, 3, 4. Is it possible to have a graph with degrees 2, 2, 2, 2, 4? Show that it is possible by drawing such a graph or prove it's not possible.

(f) The degrees of the nodes in the above example is 2, 2, 3, 3, 4. Is it possible to have a graph with degrees 2, 2, 3, 3, 3? Show that it is possible by drawing such a graph or prove it's not possible.

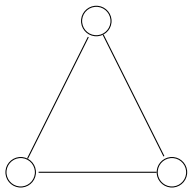
(g) Is it possible to have a triangulation with no nodes of degree 2 or 3?

(h) Suppose you have a triangular graph G with a node v of degree 2. If there are at least 4 nodes in G, after removing v from G and removing all the edges attached to v, is the resulting graph a triangulation graph? Prove that this is true or provide a counterexample.

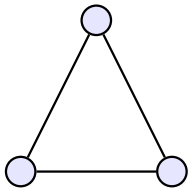
SOLUTION.

How to draw graphs using my library

This is from the previous question:



Look at `main.tex` for the LaTeX code to draw the above. You can also do this:



Look at `main.tex` for the python code to draw the above.

Q11. Sipser 0.13.

SOLUTION.